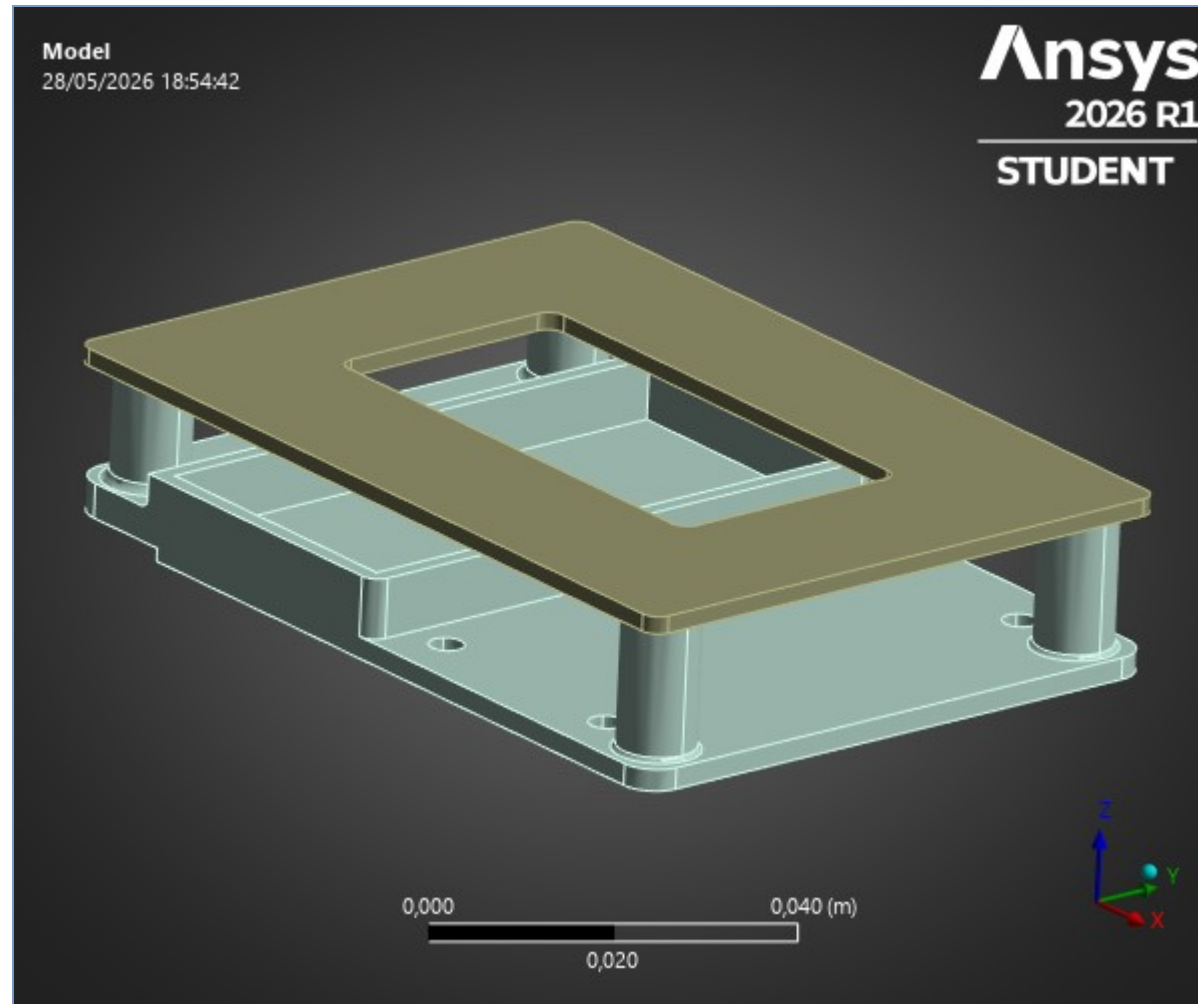




Project

First Saved	Tuesday, May 26, 2026
Last Saved	Wednesday, May 27, 2026
Product Version	2026 R1
Save Project Before Solution	No
Save Project After Solution	No



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Units

TABLE 1

Unit System	Metric (m, kg, N, s, V, A) Degrees rad/s Celsius
Angle	Degrees
Rotational Velocity	rad/s
Temperature	Celsius

Model (A4)

TABLE 2
Model (A4) > Geometry Imports

Object Name	Geometry Imports
State	Solved

TABLE 3
Model (A4) > Geometry Imports > Geometry Import (A3)

Object Name	Geometry Import (A3)
State	Solved
Definition	
Source	C:\Users\Mateus\OneDrive\Documentos\UnB\4 sem\PI\Chassi\Final chassis\chassis.stp
Type	Step
Basic Geometry Options	
Solid Bodies	Yes
Surface Bodies	Yes
Line Bodies	No
Parameters	Independent
Parameter Key	ANS;DS
Attributes	No
Named Selections	No
Material Properties	No
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	No
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Mixed Import Resolution	None
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	Program Tolerance
Stitch Tolerance	0,0000001
Decompose Disjoint Geometry	Yes

Enclosure and Symmetry Processing	Yes
-----------------------------------	-----

Geometry

TABLE 4
Model (A4) > Geometry

Object Name	Geometry
State	Fully Defined
Definition	
Source	C:\Users\Mateus\OneDrive\Documentos\UnB\4 sem\PI\Chassi\Final chassis\chassis.stp
Type	Step
Length Unit	Millimeters
Element Control	Program Controlled
Display Style	Body Color
Bounding Box	
Length X	0,1 m
Length Y	7,e-002 m
Length Z	2,05e-002 m
Properties	
Volume	3,4711e-005 m ³
Mass	4,3041e-002 kg
Scale Factor Value	1,
Statistics	
Bodies	2
Active Bodies	2
Nodes	6897
Elements	3228
Mesh Metric	None
Update Options	
Assign Default Material	No
Basic Geometry Options	
Solid Bodies	Yes
Surface Bodies	Yes
Line Bodies	No
Parameters	Independent
Parameter Key	ANS;DS
Attributes	No

Named Selections	No
Material Properties	No
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	No
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Mixed Import Resolution	None
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	Program Tolerance
Decompose Disjoint Geometry	Yes
ID_GeometryPrefProcessPhysicsDefinition	No
Enclosure and Symmetry Processing	Yes

TABLE 5
Model (A4) > Geometry > Parts

Object Name	Part2 PartBody	1st level PartBody
State	Meshed	
Graphics Properties		
Visible	Yes	
Transparency	1	
Definition		
Suppressed	No	
Stiffness Behavior	Flexible	
Coordinate System	Default Coordinate System	
Reference Temperature	By Environment	
Treatment	None	
Material		
Assignment	PLA	
Nonlinear Effects	Yes	
Thermal Strain Effects	Yes	
Bounding Box		
Length X	0,1 m	

Length Y	7,e-002 m	
Length Z	7,2e-003 m	1,85e-002 m
Properties		
Volume	1,0729e-005 m ³	2,3981e-005 m ³
Mass	1,3304e-002 kg	2,9737e-002 kg
Centroid X	-5,e-003 m	-8,959e-003 m
Centroid Y	3,9581e-014 m	4,5509e-013 m
Centroid Z	1,939e-002 m	3,2278e-003 m
Moment of Inertia Ip1	7,0995e-006 kg·m ²	1,4703e-005 kg·m ²
Moment of Inertia Ip2	1,3915e-005 kg·m ²	2,8909e-005 kg·m ²
Moment of Inertia Ip3	2,0994e-005 kg·m ²	4,2853e-005 kg·m ²
Statistics		
Nodes	2115	4782
Elements	925	2303
Mesh Metric	None	

TABLE 6
Model (A4) > Materials

Object Name	<i>Materials</i>
State	Fully Defined
Statistics	
Materials	2
Material Assignments	0

Coordinate Systems

TABLE 7
Model (A4) > Coordinate Systems > Coordinate System

Object Name	<i>Global Coordinate System</i>
State	Fully Defined
Definition	
Type	Cartesian
Coordinate System ID	0,
Origin	
Origin X	0, m
Origin Y	0, m
Origin Z	0, m

Directional Vectors	
X Axis Data	[1, 0, 0,]
Y Axis Data	[0, 1, 0,]
Z Axis Data	[0, 0, 1,]
Transfer Properties	
Source	
Read Only	No

Connections

TABLE 8
Model (A4) > Connections

Object Name	Connections
State	Fully Defined
Auto Detection	
Generate Automatic Connection On Refresh	Yes
Transparency	
Enabled	Yes
Statistics	
Contacts	1
Active Contacts	1
Joints	0
Active Joints	0
Beams	0
Active Beams	0
Bearings	0
Active Bearings	0
Springs	0
Active Springs	0
Body Interactions	0
Active Body Interactions	0

TABLE 9
Model (A4) > Connections > Contacts

Object Name	Contacts
State	Fully Defined
Definition	

Connection Type	Contact
Scope	
Scoping Method	Geometry Selection
Geometry	All Bodies
Auto Detection	
Tolerance Type	Slider
Tolerance Slider	0,
Tolerance Value	3,0944e-004 m
Use Range	No
Face/Face	Yes
Face-Face Angle Tolerance	75, °
Face Overlap Tolerance	Off
Cylindrical Faces	Include
Face/Edge	No
Edge/Edge	No
Priority	Include All
Group By	Bodies
Search Across	Bodies
Statistics	
Connections	1
Active Connections	1

TABLE 10
Model (A4) > Connections > Contacts > Contact Regions

Object Name	<i>Contact Region</i>
State	Fully Defined
Scope	
Scoping Method	Geometry Selection
Contact	17 Faces
Target	12 Faces
Contact Bodies	Part2 PartBody
Target Bodies	1st level PartBody
Protected	No
Definition	
Type	Bonded
Scope Mode	Automatic
Behavior	Program Controlled

Trim Contact	Program Controlled
Trim Tolerance	3,0944e-004 m
Contact APDL Name	
Target APDL Name	
Suppressed	No
Display	
Element Normals	No
Advanced	
Formulation	Program Controlled
Small Sliding	Program Controlled
Detection Method	Program Controlled
Penetration Tolerance	Program Controlled
Elastic Slip Tolerance	Program Controlled
Normal Stiffness	Program Controlled
Update Stiffness	Program Controlled
Pinball Region	Program Controlled
Geometric Modification	
Contact Geometry Correction	None
Target Geometry Correction	None

Mesh

TABLE 11
Model (A4) > Mesh

Object Name	<i>Mesh</i>
State	Solved
Display	
Display Style	Use Geometry Setting
Defaults	
Physics Preference	Mechanical
Element Order	Program Controlled
Element Size	Default (6,1887e-003 m)
Sizing	
Mesh Sizing	Uniform
Bounding Box Diagonal	0,12377 m
Average Surface Area	3,1354e-004 m ²
Minimum Edge Length	3,1416e-004 m

Quality	
Check Mesh Quality	Yes, Errors
Error Limits	Aggressive Mechanical
Target Element Quality	Default (0,1)
Smoothing	Medium
Mesh Optimization	No
Mesh Metric	None
Inflation	
Use Automatic Inflation	None
Inflation Option	Smooth Transition
Transition Ratio	0,272
Maximum Layers	5
Growth Rate	1,2
Inflation Algorithm	Pre
Inflation Element Type	Wedges
View Advanced Options	No
Advanced	
Number of CPUs for Parallel Part Meshing	Program Controlled
Straight Sided Elements	No
Rigid Body Behavior	Dimensionally Reduced
Triangle Surface Mesher	Program Controlled
Topology Checking	Yes
Pinch Tolerance	Default (5,5699e-004 m)
Generate Pinch on Refresh	No
Auto-Map Fillets	No
Automatic Methods	
Sheet Body Method	Prime Quad Dominant
Sweepable Body Method	Sweep
Statistics	
Nodes	6897
Elements	3228
Show Detailed Statistics	No

Static Structural (A5)

TABLE 12
Model (A4) > Analysis

Object Name	<i>Static Structural (A5)</i>
State	Solved
Definition	
Physics Type	Structural
Analysis Type	Static Structural
Solver Target	Mechanical APDL
Options	
Environment Temperature	22, °C
Generate Input Only	No

TABLE 13
Model (A4) > Static Structural (A5) > Analysis Settings

Object Name	<i>Analysis Settings</i>
State	Fully Defined
Step Controls	
Number Of Steps	1,
Current Step Number	1,
Step End Time	1, s
Auto Time Stepping	Program Controlled
Maximum Iterations Per Substep	Program Controlled
Solver Controls	
Solver Type	Program Controlled
Weak Springs	Off
Solver Pivot Checking	Program Controlled
Large Deflection	Off
Inertia Relief	Off
Quasi-Static Solution	Off
Rotordynamics Controls	
Coriolis Effect	Off
Restart Controls	
Generate Restart Points	Program Controlled
Retain Files After Full Solve	No
Combine Restart Files	Program Controlled
Nonlinear Controls	
Newton-Raphson Option	Program Controlled
Force Convergence	Program Controlled
Moment Convergence	Program Controlled

Displacement Convergence	Program Controlled
Rotation Convergence	Program Controlled
Line Search	Program Controlled
Stabilization	Program Controlled
Advanced	
Inverse Option	No
Contact Split (DMP)	Program Controlled
Output Controls	
Output Selection	None
Stress	Yes
Back Stress	No
Strain	Yes
Contact Data	Yes
Nonlinear Data	No
Nodal Forces	No
Volume and Energy	Yes
Euler Angles	Yes
General Miscellaneous	No
Contact Miscellaneous	No
Store Results At	All Time Points
Result File Options	
Result File Compression	Program Controlled
Analysis Data Management	
Solver Files Directory	C:\Users\Mateus\OneDrive\Documentos\UnB\4 sem\PI\Chassi\chassi ansys_files\dp0\SYS\MECH\
Future Analysis	None
Scratch Solver Files Directory	
Save MAPDL db	No
Contact Summary	Program Controlled
Delete Unneeded Files	Yes
Nonlinear Solution	No
Solver Units	Active System
Solver Unit System	mks

TABLE 14
Model (A4) > Static Structural (A5) > Accelerations

Object Name	<i>Acceleration</i>
State	Fully Defined

Scope	
Geometry	All Bodies
Definition	
Define by	Vector
Magnitude	15, m/s ² (ramped)
Direction	Defined
Suppressed	No

FIGURE 1
Model (A4) > Static Structural (A5) > Acceleration

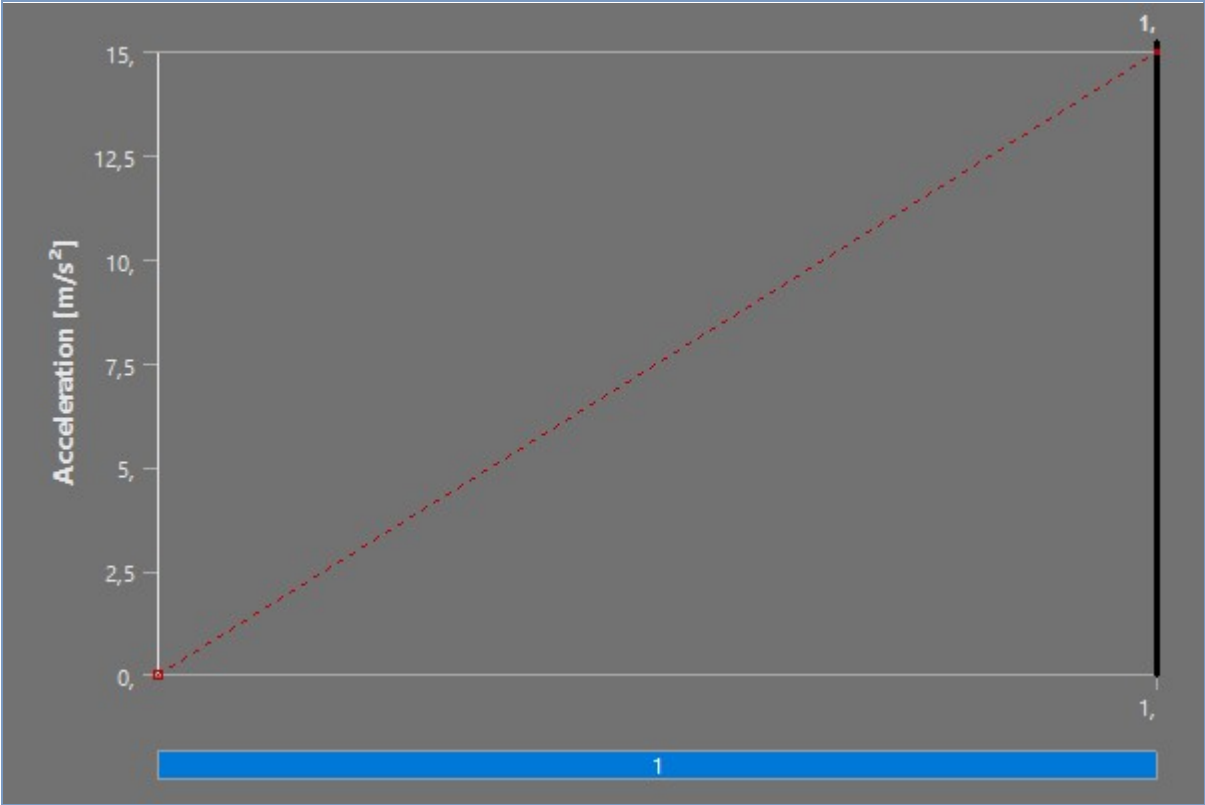


TABLE 15
Model (A4) > Static Structural (A5) > Loads

Object Name	<i>Displacement</i>	<i>Fixed Support</i>	<i>Force</i>	<i>Force 2</i>
-------------	---------------------	----------------------	--------------	----------------

State	Fully Defined		
Scope			
Scoping Method	Geometry Selection		
Geometry	2 Faces	8 Faces	1 Face
Definition			
Type	Displacement	Fixed Support	Force
Define by	Components		Vector
Coordinate System	Global Coordinate System		
X Component	Free		
Y Component	Free		
Z Component	0, m (ramped)		
Suppressed	No		
Applied By			Surface Effect
Magnitude			0,5635 N (ramped) 0,98 N (ramped)
Direction			Defined

FIGURE 2
Model (A4) > Static Structural (A5) > Displacement

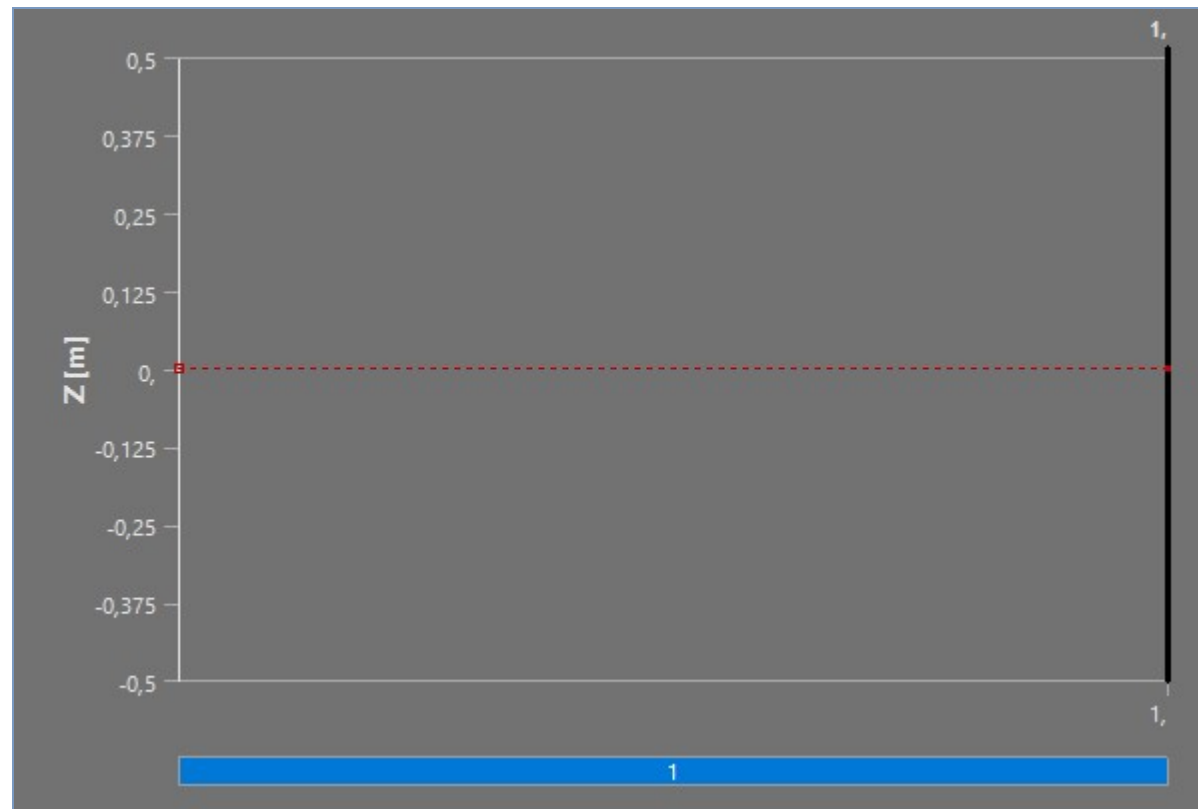


FIGURE 3
Model (A4) > Static Structural (A5) > Force

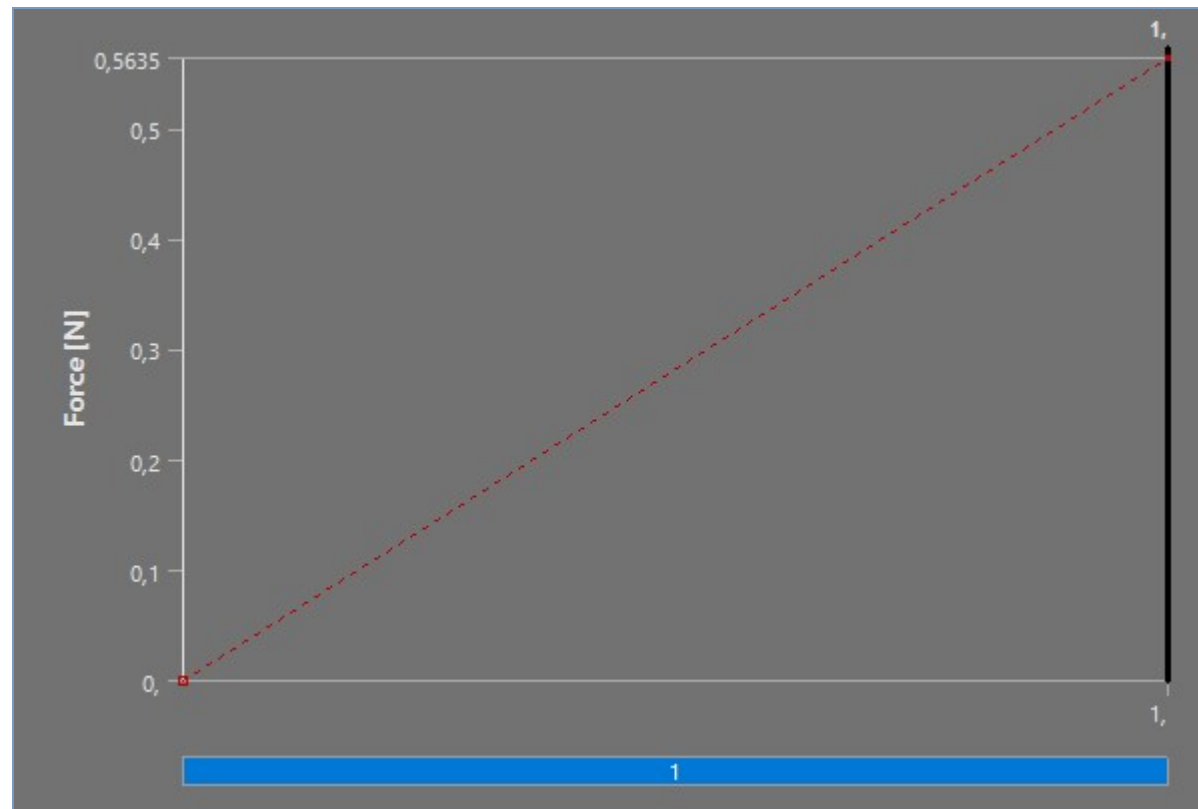
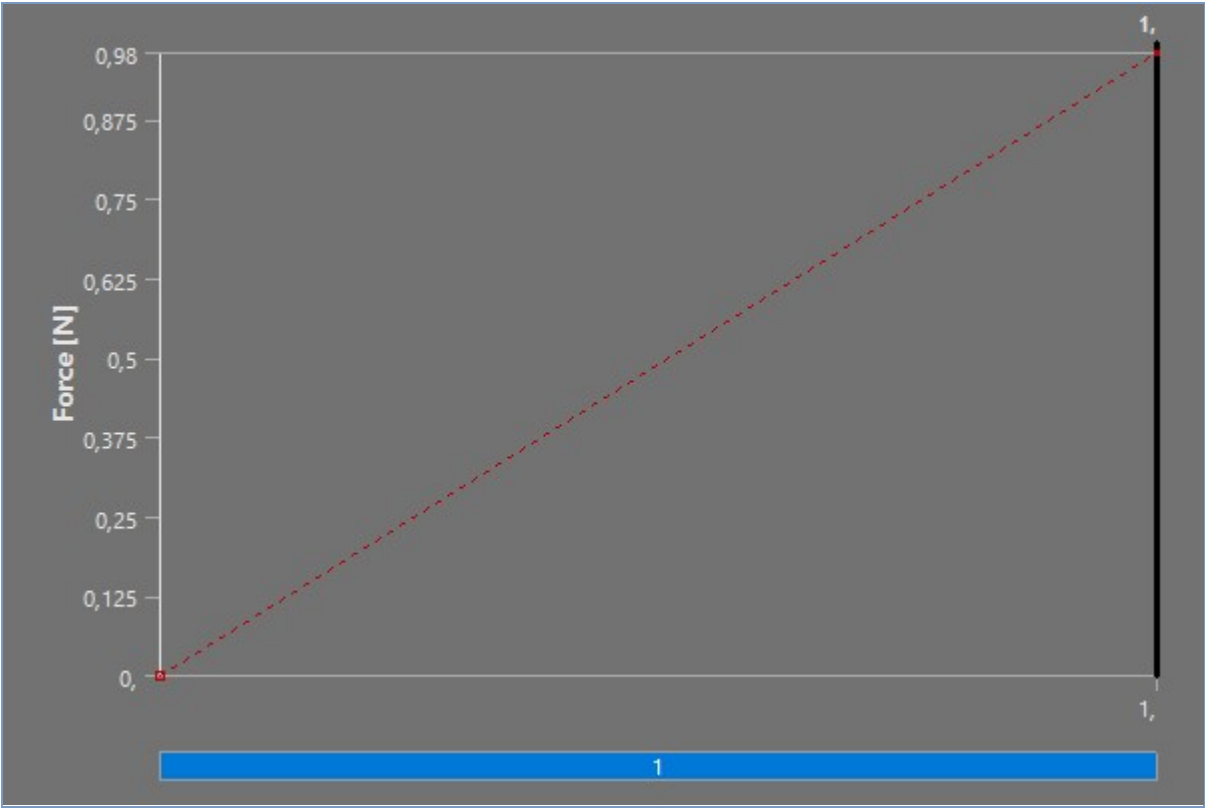


FIGURE 4
Model (A4) > Static Structural (A5) > Force 2



Solution (A6)

TABLE 16
Model (A4) > Static Structural (A5) > Solution

Object Name	<i>Solution (A6)</i>
State	Solved
Adaptive Mesh Refinement	
Max Refinement Loops	1,
Refinement Depth	2,
Information	
Status	Done
MAPDL Elapsed Time	17,205 s
MAPDL Memory Used	218, MB

MAPDL Result File Size	2,5 MB
Post Processing	
Beam Section Results	No
On Demand Stress/Strain	No

TABLE 17
Model (A4) > Static Structural (A5) > Solution (A6) > Solution Information

Object Name	<i>Solution Information</i>
State	Solved
Solution Information	
Solution Output	Solver Output
Newton-Raphson Residuals	0
Identify Element Violations	0
Update Interval	2,5 s
Display Points	All
FE Connection Visibility	
Activate Visibility	Yes
Display	All FE Connectors
Draw Connections Attached To	All Nodes
Line Color	Connection Type
Visible on Results	No
Line Thickness	Single
Display Type	Lines

TABLE 18
Model (A4) > Static Structural (A5) > Solution (A6) > Results

Object Name	Total Deformation	Equivalent Stress
State	Solved	
Scope		
Scoping Method	Geometry Selection	
Geometry	All Bodies	
Definition		
Type	Total Deformation	Equivalent (von-Mises) Stress
By	Time	
Display Time	Last	
Separate Data by Entity	No	
Calculate Time History	Yes	

Identifier		
Suppressed	No	
Results		
Minimum	0, m	16,755 Pa
Maximum	1,0418e-005 m	1,6446e+005 Pa
Average	1,0428e-006 m	12822 Pa
Minimum Occurs On	1st level PartBody	
Maximum Occurs On	Part2 PartBody	1st level PartBody
Information		
Time	1, s	
Load Step	1	
Substep	1	
Iteration Number	1	
Integration Point Results		
Display Option		Averaged
Average Across Bodies		No

FIGURE 5
Model (A4) > Static Structural (A5) > Solution (A6) > Total Deformation

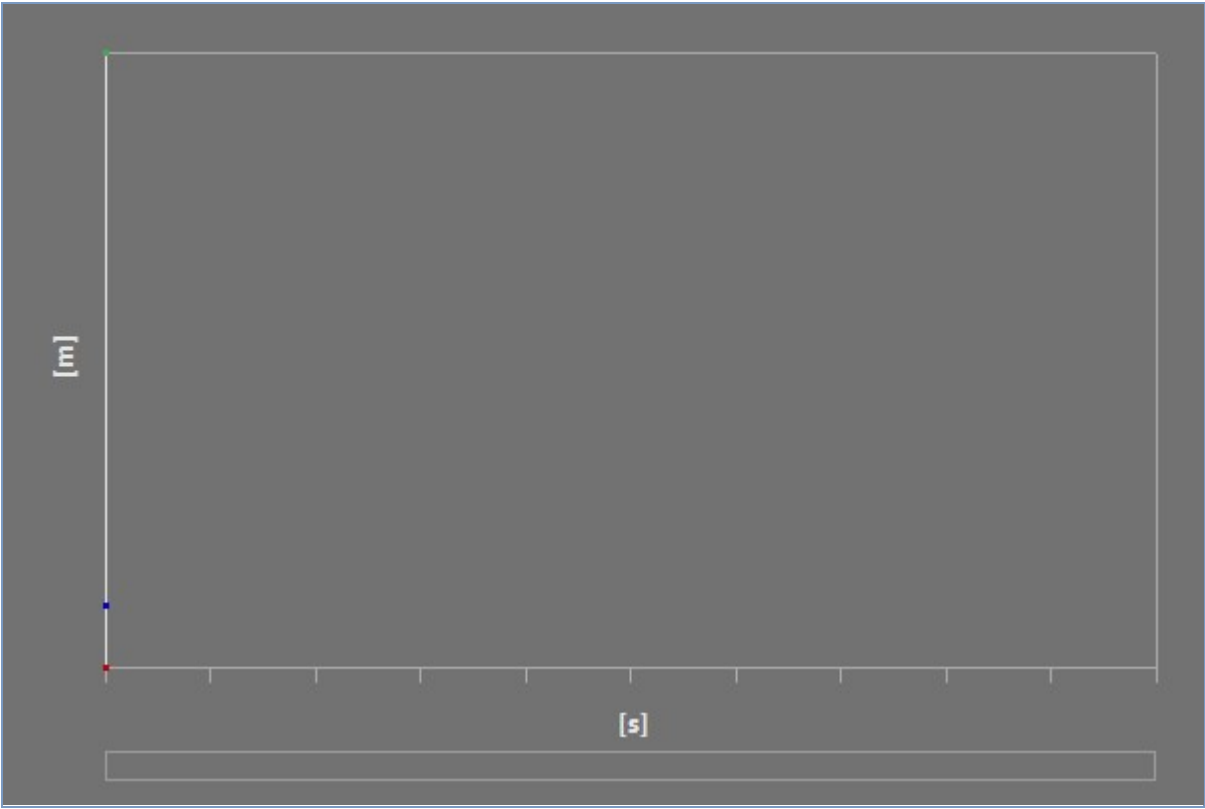


TABLE 19
Model (A4) > Static Structural (A5) > Solution (A6) > Total Deformation

Time [s]	Minimum [m]	Maximum [m]	Average [m]
1,	0,	1,0418e-005	1,0428e-006

FIGURE 6
Model (A4) > Static Structural (A5) > Solution (A6) > Equivalent Stress

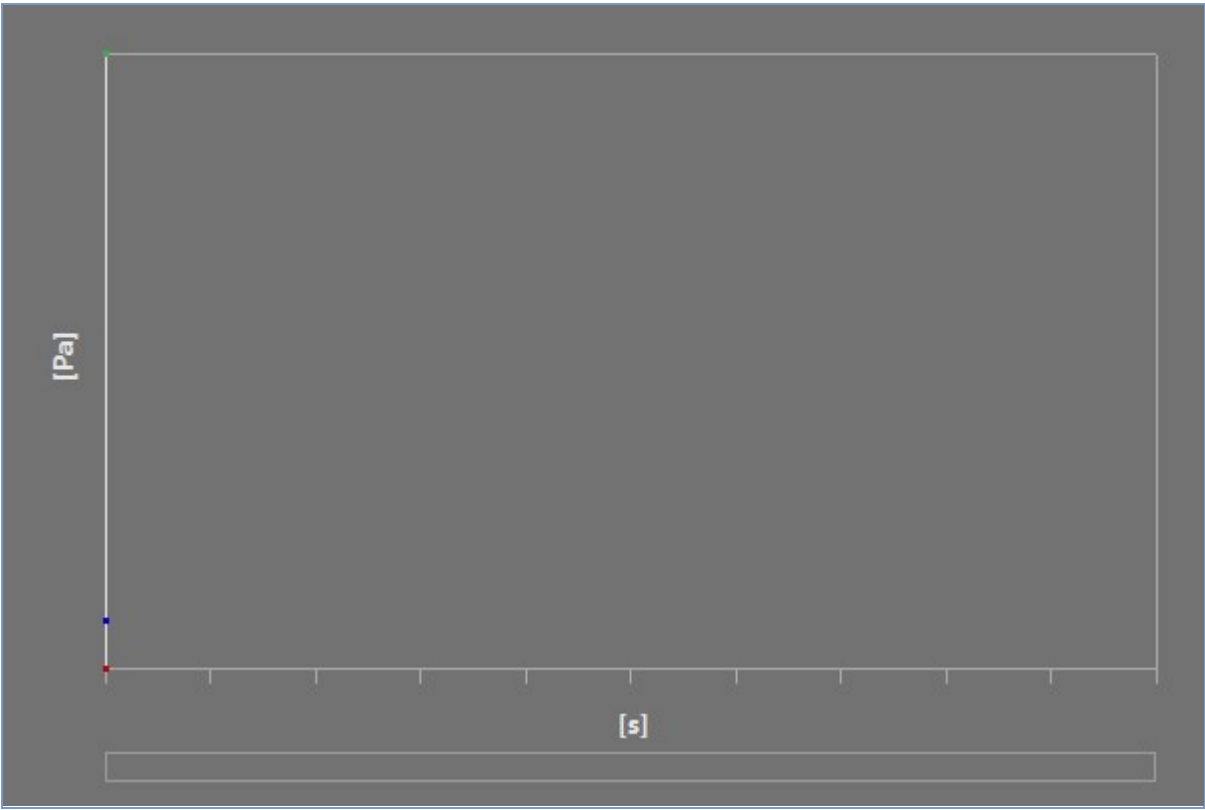


TABLE 20
Model (A4) > Static Structural (A5) > Solution (A6) > Equivalent Stress

Time [s]	Minimum [Pa]	Maximum [Pa]	Average [Pa]
1,	16,755	1,6446e+005	12822

Material Data

PLA

TABLE 21
PLA > Constants

Density	1240, kg m ⁻³
---------	--------------------------

TABLE 22
PLA > Color

Red	Green	Blue
235,	222,	222,

TABLE 23
PLA > Isotropic Elasticity

Young's Modulus Pa	Poisson's Ratio	Bulk Modulus Pa	Shear Modulus Pa	Temperature C
3,5e+009	0,36	4,1667e+009	1,2868e+009	

TABLE 24
PLA > Tensile Yield Strength

Tensile Yield Strength Pa
5,e+007